MDRS SC-32 BAHADDURGHATTA (KOGUNDE)

Chitradurga Taluk & District — 577519

Student Notes

Level 3 — Classes 8, 9, 10

ICT Workbook - Level 3

Name: _____

Class: _____ Section: _____

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How to Use This Workbook

Write in pencil or pen. Complete Try It before checking answers. Use Check Yourself to test yourself. Fill in Today I Learned after every lesson. Use the Revision Page before each assessment. The Glossary is at the back.

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Chapter Map — Advanced Programming Concepts

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- 1.11 Introduction to Recursion
- 1.12 Programming Project

Key terms to know:

Lesson 1.01 Functions Introduction

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze how a function and identify its components: name, parameters, body] [- I can write a simple function with no parameters and call it from mAIn program] [- I can explAIn why functions help avoid code repetition and improve readability]

Key Words

Function	Reusable block of code that performs one specific task perfectly each time
Definition	Writing function code once so it can be called later anywhere
Call	Telling program to run the function by writing its name with brackets
DRY	Do Not Repeat Yourself principle encouraging reuse of code efficiently

Core Idea

A function in Python is a named block of code that performs a specific task.

Sketch the diagram or concept here.

Worked Example

Step 1: I wrote the morning_assembly function and called it three times.
Step 2: Then I created lunch_routine with wash hands, collect food, eat quietly, return plate.

Try It (write your answers)

1. List or name: Analyze how a function and its components: name, parameters, body.

Answer: See Key Words and Core Idea above.

2. Give one real-life example of Functions Introduction.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.02 Function Parameters

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze the difference between parameters and arguments and distinguish between the two terms] [- I can write functions that accept one or more parameters and pass values when calling] [- I can explAIn how parameters make functions flexible and reusable with different data]

Key Words

Parameter	Variable listed in function definition receiving value when called correctly for flexibility
Argument	Actual value you pass to function when you call it for execution
Input	Data given to function so it works with different values
Signature	First line showing function name and its expected parameters clearly together

Core Idea

A parameter is a variable listed in the function definition. It is a placeholder that holds the value you pass to the function.

Sketch the diagram or concept here.

Worked Example

Step 1: My calculate_area function works with any two numbers.
Step 2: I tested it with classroom dimensions 8 meters by 6 meters and got 48 square meters.

Try It (write your answers)

1. Explain: Analyze the difference between parameters and arguments and distinguish between the two terms.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Function Parameters.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.03 Return Values

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use the return keyword to send a result back from a function to the calling code] [- I can store the returned value in a variable and use it in further calculations] [- I can compare functions that print results versus functions that return results]

Key Words

Return	Sending result back from function to the place that called it
Value	Result produced by function after doing its calculation correctly for use
Caller	Code that calls function and waits to receive returned result back
None	Special value meaning function returned nothing useful for caller to use

Core Idea

The return statement ends the function execution and sends a value back to the caller. The returned value can be stored in a variable for later use.

Sketch the diagram or concept here.

Worked Example

Step 1: My convert_temperature function takes 100 Celsius and returns 212 Fahrenheit.
Step 2: My calculate_grade function takes 85 and returns A, takes 65 and returns B, and takes 45 and returns C.

Try It (write your answers)

1. Explain: Use the return keyword to send a result back from a function to the calling code.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Return Values.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.04 Arrays Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can create and manipulate a list in Python and understand that lists can hold multiple values in one variable] [- I can access individual elements using index numbers starting from zero] [- I can modify list elements by assigning new values to specific index positions]

Key Words

Array	Ordered list storing many values of same type under one name
Index	Number showing position of element starting from zero in array quickly
Element	Single value stored at one position inside array for access later
Size	Total number of elements currently stored inside the array altogether

Core Idea

A list in Python is created using square brackets with values separated by commas. Each element has an index starting from zero.

Sketch the diagram or concept here.

Worked Example

Step 1: My attendance list has present for six students and absent for one.
Step 2: My subjects list has Mathematics Science English Kannada and Social Studies.

Try It (write your answers)

1. Explain: Create and manipulate a list in Python and understand that lists can hold multiple values in one variable.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Arrays Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.05 Array Operations

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use append and insert to add elements to a list at specified positions] [- I can use remove and pop to delete elements from a list] [- I can use the len function and the in operator to check list properties]

Key Words

Traverse	Visiting each element of array one by one in order sequentially
Insert	Adding new element at specific position inside existing array carefully
Delete	Removing element from array and shifting others to fill gap
Search	Looking through array to find element matching desired value quickly

Core Idea

The append method adds an element to the end of the list. The insert method adds an element at a specific index position.

Sketch the diagram or concept here.

Worked Example

Step 1: My roster started empty.
Step 2: I added Arjun Priya Ravi Kavya and Meena.

Try It (write your answers)

1. Explain: Use append and insert to add elements to a list at specified positions.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Array Operations.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.06 String Operations

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can perform string concatenation and repetition to combine and repeat text] [- I can use string methods like upper lower strip and replace to transform text] [- I can slice strings to extract substrings using index ranges]

Key Words

String	Sequence of characters like name or sentence inside quotes together
Concatenation	Joining two strings together end to end to make one longer string
Substring	Part of a string taken from middle like first three letters
Length	Number of characters including spaces inside a string counted together

Core Idea

String concatenation uses the plus operator to join two strings. String repetition uses the multiply operator to repeat a string multiple times.

Sketch the diagram or concept here.

Worked Example

Step 1: My ID card shows our school ACADEMY in uppercase at the top.
Step 2: The student name is Priya Sharma and her class is Class 9 Section A.

Try It (write your answers)

1. Explain: Perform string concatenation and repetition to combine and repeat text.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of String Operations.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.07 Nested Loops

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can write a loop inside another loop to process multi-dimensional data patterns] [- I can predICT the output of nested loops by tracing through iterations step by step] [- I can apply nested loops to solve real problems like multiplication tables and grid patterns]

Key Words

Nested Loop	Loop placed inside another loop running inner many times per outer
Outer Loop	First loop controlling how many times inner repeats completely each cycle
Inner Loop	Second loop running fully each time outer takes one step forward
Iteration	Single execution of loop body before checking condition again for next

Core Idea

A nested loop is a loop inside another loop. The outer loop controls the number of times the inner loop repeats entirely.

Sketch the diagram or concept here.

Worked Example

Step 1: My multiplication table prints neatly in columns.
Step 2: The triangle program uses the outer loop for rows and the inner loop prints j stars for row j.

Try It (write your answers)

1. Explain: Write a loop inside another loop to process multi-dimensional data patterns.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Nested Loops.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.08 Exception Handling

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze and compare common Python exceptions including TypeError ValueError and ZeroDivisionError] [- I can use try except blocks to handle errors gracefully without crashing the program] [- I can write programs that provide meaningful error messages to guide the user]

Key Words

Exception	Error happening during program run that would stop execution if unhandled
Try	Block where you attempt code that might cause an error safely
Except	Block running alternative code when try block fails with error gracefully
Finally	Block always running whether error happened or not for cleanup

Core Idea

An exception is an error that occurs during program execution.

Sketch the diagram or concept here.

Worked Example

Step 1: My calculator catches division by zero and shows Please enter a non-zero divisor.
Step 2: It catches invalid input and shows Please enter valid numbers.

Try It (write your answers)

1. Explain: Analyze and compare common Python exceptions including TypeError ValueError and ZeroDivisionError.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Exception Handling.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.09 File Input and Output

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can open read and close text files using the open function and with statement] [- I can write data to a file using write and writelines methods] [- I can use file handling to save and retrieve program data persistently]

Key Words

File	Collection of data stored on disk with a name like notes.txt
Read	Getting data from file into program for processing or display on screen
Write	Saving data from program into file for future use permanently
Path	Address showing where file is stored like folder inside folder

Core Idea

File handling in Python uses the open function with a filename and mode. The write mode w creates a new file or overwrites an existing one.

Sketch the diagram or concept here.

Worked Example

Step 1: My report card file has the student name class and marks formatted nicely.
Step 2: My results program reads the marks calculates total as 235 and average as 78.33 and saves it.

Try It (write your answers)

1. Explain: Open read and close text files using the open function and with statement.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of File Input and Output.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.10 Modular Programming

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can break a large program into smaller functions each handling a specific task] [- I can understand how modules help organize code and make it reusable] [- I can import and use built-in modules like math random and datetime]

Key Words

Module	File containing related functions that can be reused in many programs
Import	Bringing module code into current program to use its functions
Reuse	Using same code in many places without writing again for efficiency
Cohesion	How well parts inside one module work together for single purpose

Core Idea

Modular programming means organizing code into separate self-contained units. Each unit handles one specific task.

Sketch the diagram or concept here.

Worked Example

Step 1: My program has four functions each handling one task.
Step 2: The main menu loops and lets the user choose an option.

Try It (write your answers)

1. Explain: Break a large program into smaller functions each handling a specific task.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Modular Programming.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.11 Introduction to Recursion

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze recursion as a function that calls itself with a modified parameter and justify your choice] [- I can write a recursive function with a clear base case to stop the recursion] [- I can trace through a recursive function call to understand the call stack]

Key Words

Recursion	Function calling itself to solve smaller version of same problem cleverly
Base Case	Simple situation where recursion stops and returns answer directly without further calls
Recursive Call	Function calling itself with smaller input to move toward base case gradually
Stack	Memory storing each waiting function call until base case reached for completion

Core Idea

Recursion is when a function calls itself. Every recursive function needs two parts. The base case is the condition that stops the recursion.

Sketch the diagram or concept here.

Worked Example

Step 1: My factorial function returns 120 for factorial of 5 because 5 times 4 times 3 times 2 times 1 equals 120.
Step 2: factorial of 0 returns 1 which is the base case.

Try It (write your answers)

1. Explain: Analyze recursion as a function that calls itself with a modified parameter and justify your choice.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Introduction to Recursion.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 1.12 Programming Project

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can apply functions parameters return values lists and file handling in a complete project] [- I can plan a program by identifying functions needed and how they connect together] [- I can debug and test a multi-function program to ensure all parts work correctly]

Key Words

Programming	Writing instructions so computer performs tasks step by step
Code	Written instructions that computer can understand and follow exactly
Instruction	Single command telling computer to do one specific action
Bug	Mistake in code that makes program behave incorrectly or stop

Core Idea

A programming project requires planning before coding. Start by identifying all the functions your program needs.

Sketch the diagram or concept here.

Worked Example

Step 1: My library system has six functions: add_book issue_book return_book display_books search_book and load_data.
Step 2: I use a list of dictionaries where each dictionary represents a book.

Try It (write your answers)

1. Explain: Apply functions parameters return values lists and file handling in a complete project.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Programming Project.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Advanced Programming Concepts

1.01 I will give you a short program that has repeated code
1.02 I will give you a function definition with two parameters
1.03 I will give you a problem description
1.04 I will give you a list and ask you questions about specific elements using…
1.05 I will give you a sequence of operations to perform on a list
1.06 I will give you a string and ask you to perform specific operations on it
1.07 I will give you a nested loop program and ask you to trace through it completely
1.08 I will give you programs with potential errors
1.09 I will give you a scenario where a program needs to save and load data
1.10 I will give you a monolithic program with everything in one function
1.11 I will give you a recursive function and ask you to trace through it completely
1.12 I will evaluate your project based on these criteria: correct use of functions…

Key terms: Function, Definition, Call, DRY, Parameter, Argument, Input, Signature, Return, Value

I still need help with:

Chapter Map — Data Structures

- 2.01 Lists Advanced
- 2.02 List Methods
- 2.03 Dictionaries
- 2.04 Dictionary Operations
- 2.05 Tuples
- 2.06 Sets
- 2.07 Nested Structures
- 2.08 Sorting Algorithms
- 2.09 Searching Algorithms
- 2.10 Stacks and Queues
- 2.11 Data Structure Selection
- 2.12 Data Structure Project

Key terms to know:

Lesson 2.01 Lists Advanced

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use list slicing to extract sublists using start stop and step parameters] [- I can apply list comprehension to create lists from expressions concisely] [- I can understand list copying and the difference between shallow and reference copies]

Key Words

List	Ordered collection holding many values that can change during program run
Index	Position number starting zero indicating where element lives in list currently
Slice	Part of list taken as new smaller list like first three items
Mutability	Ability to change list contents after creation without making new list

Core Idea

List slicing uses the syntax list start colon stop colon step. Start is where slicing begins inclusive. Stop is where it ends exclusive.

Sketch the diagram or concept here.

Worked Example

Step 1: My slices give first 5 marks from index 0 to 4 and last 5 from index 15 to 19.
Step 2: Every third student uses step 3.

Try It (write your answers)

1. Explain: Use list slicing to extract sublists using start stop and step parameters.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Lists Advanced.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.02 List Methods

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use sort and reverse to order list elements in ascending and descending order] [- I can apply count and index methods to find element frequency and position] [- I can use extend and clear methods to merge lists and empty them completely]

Key Words

Method	Function belonging to list called with dot like list.append for action
Append	Adding item to end of list without changing other elements quickly
Pop	Removing and returning last item from list for use elsewhere immediately
Sort	Arranging list elements in order like smallest to largest automatically quickly

Core Idea

The sort method arranges elements in ascending order by default. Pass reverse equals true to sort in descending order.

Sketch the diagram or concept here.

Worked Example

Step 1: My sorted list shows lowest 35 and highest 98.
Step 2: Eight students scored above 80.

Try It (write your answers)

1. Explain: Use sort and reverse to order list elements in ascending and descending order.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of List Methods.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.03 Dictionaries

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can create a dICTionary with key-value pAIrs and understand how keys work] [- I can access modify add and delete dICTionary entries using key names] [- I can iterate through dICTionaries using keys values and items methods]

Key Words

Dictionary	Collection storing pairs of key and value for quick lookup by name
Key	Unique name used to find its matching value inside dictionary quickly
Value	Data stored under a key that you retrieve when searching dictionary
Pair	Single key together with its value as one entry inside dictionary

Core Idea

A dictionary stores data as key-value pairs. Keys must be unique and immutable like strings numbers or tuples. Values can be any data type.

Sketch the diagram or concept here.

Worked Example

Step 1: My dictionary has name Arya, class 9, roll 15, subjects list with 5 subjects and marks list with 5 marks.
Step 2: I accessed name directly.

Try It (write your answers)

1. Explain: Create a dictionary with key-value pairs and understand how keys work.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Dictionaries.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.04 Dictionary Operations

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use get method to safely access dICTionary values with default values] [- I can check if a key exists using the in operator before accessing it] [- I can merge dICTionaries using update method and understand dICTionary comprehensions]

Key Words

Lookup	Finding value by giving its key to dictionary quickly without scanning
Update	Changing value already stored under existing key to new value
Delete	Removing a key and its value from dictionary completely thereafter for cleanup
Iterate	Visiting each key-value pair one by one to process them all sequentially

Core Idea

The get method provides safe dictionary access. It takes a key and an optional default value. If the key exists it returns the value.

Sketch the diagram or concept here.

Worked Example

Step 1: My attendance dictionary has student names and their daily records.
Step 2: get returns 0 for new students.

Try It (write your answers)

1. Explain: Use get method to safely access dictionary values with default values.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Dictionary Operations.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.05 Tuples

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can create tuples and understand that they are immutable and cannot be changed] [- I can access tuple elements using indexing and slicing similar to lists] [- I can use tuples as dICTionary keys and for function multiple return values]

Key Words

Tuple	Ordered collection that cannot be changed after creation keeping data safe permanently
Immutable	Cannot be altered after creation protecting data from accidental change completely
Index	Position number starting zero used to fetch element from tuple quickly
Packing	Putting many values together into one tuple in single step conveniently

Core Idea

Tuples are ordered collections enclosed in parentheses or just commas. They are immutable meaning once created they cannot be modified.

Sketch the diagram or concept here.

Worked Example

Step 1: My coordinates tuple 12.9716 comma 77.5946 is the key for school name our school.
Step 2: My function returns the tuple with three values.

Try It (write your answers)

1. Explain: Create tuples and understand that they are immutable and cannot be changed.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Tuples.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.06 Sets

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can create sets and understand they store unique unordered elements] [- I can perform set operations including union intersection and difference] [- I can use sets to remove duplicates from a list and test membership efficiently]

Key Words

Set	Collection holding only unique values with no duplicates allowed automatically for uniqueness
Unique	Appearing only once with no repetition inside the collection for clarity
Union	Combining all elements from two sets together without duplicating any value
Intersection	Keeping only elements present in both sets together as common part

Core Idea

Sets are unordered collections of unique elements enclosed in curly braces. Duplicates are automatically removed.

Sketch the diagram or concept here.

Worked Example

Step 1: My attendance list had Arjun appearing three times.
Step 2: Converting to a set gave unique names.

Try It (write your answers)

1. Explain: Create sets and understand they store unique unordered elements.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Sets.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.07 Nested Structures

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can create lists of lists to represent 2D data like tables and matrices] [- I can create dICTionaries with list values and lists with dICTionary values] [- I can access and modify elements in nested data structures using multiple indices]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Nested structures are data structures that contain other data structures. A list of lists creates a 2D grid where each inner list is a row.

Sketch the diagram or concept here.

Worked Example

Step 1: My register has 10 student dictionaries each with name roll and marks list.
Step 2: I calculated totals by summing each student marks list and found the highest.

Try It (write your answers)

1. Explain: Create lists of lists to represent 2D data like tables and matrices.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Nested Structures.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

_____

Lesson 2.08 Sorting Algorithms

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can implement bubble sort by repeatedly comparing and swapping adjacent elements] [- I can implement selection sort by finding the minimum and placing it at the beginning] [- I can compare the efficiency of different sorting algorithms using step counting]

Key Words

Sorting	Arranging items in order like smallest to largest for easy searching later
Bubble Sort	Repeatedly swapping neighbours that are out of order until list sorted completely
Comparison	Checking which of two items is smaller to decide swapping during sort
Complexity	Measure of how many steps sorting needs as list grows bigger gradually

Core Idea

Bubble sort works by repeatedly stepping through the list comparing adjacent elements and swapping them if they are in wrong order.

Sketch the diagram or concept here.

Worked Example

Step 1: My bubble sort did 45 comparisons and 12 swaps.
Step 2: My selection sort did 45 comparisons but only 8 swaps.

Try It (write your answers)

1. Explain: Implement bubble sort by repeatedly comparing and swapping adjacent elements.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Sorting Algorithms.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.09 Searching Algorithms

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can implement linear search to find an element by checking each position sequentially] [- I can implement binary search to find an element by repeatedly halving the search space] [- I can compare when to use linear search versus binary search based on data properties]

Key Words

Search	Looking through data to find item matching desired value quickly and efficiently
Linear Search	Checking each item one by one from start until found sequentially
Binary Search	Repeatedly halving sorted list to find item much faster than linear
Target	Value you are trying to locate inside the collection of data currently

Core Idea

Linear search checks each element one by one from the beginning. It works on any list whether sorted or not.

Sketch the diagram or concept here.

Worked Example

Step 1: My linear search found roll 25 in 5 comparisons.
Step 2: Binary search found it in 3 comparisons.

Try It (write your answers)

1. Explain: Implement linear search to find an element by checking each position sequentially.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Searching Algorithms.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.10 Stacks and Queues

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand stack as a last-in-first-out data structure with push and pop operations] [- I can understand queue as a first-in-first-out data structure with enqueue and dequeue] [- I can apply stacks and queues to solve real-world problems like undo and printing]

Key Words

Stack	Pile where last added item is removed first like plates stacked
Queue	Line where first added item is removed first like school queue
Push	Adding new item to top of stack for storage immediately quickly
Pop	Removing top item from stack or front item from queue for use

Core Idea

A stack follows last-in-first-out principle. The last element added is the first one removed. Push adds an element to the top.

Sketch the diagram or concept here.

Worked Example

Step 1: My text editor stack stores each version of the text.
Step 2: Undo pops and restores the previous version.

Try It (write your answers)

1. Explain: Understand stack as a last-in-first-out data structure with push and pop operations.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Stacks and Queues.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.11 Data Structure Selection

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can compare lists dICTionaries sets and tuples to choose the right structure for a task] [- I can analyze the strengths and limitations of each data structure for different scenarios] [- I can apply decision criteria to select the most efficient data structure for a problem]

Key Words

Data	Facts like numbers and words stored on a computer
Raw Data	Unprocessed facts collected before cleaning or organising properly
Information	Useful meaning produced after processing raw data carefully
Dataset	Collection of related data stored together in one place

Core Idea

Lists are ordered and mutable. Use them when you need to maintain insertion order and modify elements. Dictionaries map keys to values.

Sketch the diagram or concept here.

Worked Example

Step 1: Timetable needs a dictionary with period as key and subject as value.
Step 2: Library visitors need a set for uniqueness.

Try It (write your answers)

1. Explain: Compare lists dictionaries sets and tuples to choose the right structure for a task.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Data Structure Selection.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 2.12 Data Structure Project

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can design a complete program that uses multiple data structures to solve a problem] [- I can implement lists dICTionaries and other structures to manage real data] [- I can document the design choices and explAIn why each data structure was selected]

Key Words

Data	Facts like numbers and words stored on a computer
Raw Data	Unprocessed facts collected before cleaning or organising properly
Information	Useful meaning produced after processing raw data carefully
Dataset	Collection of related data stored together in one place

Core Idea

A successful project requires careful planning before coding. Identify all features your program needs.

Sketch the diagram or concept here.

Worked Example

Step 1: My project has seven functions using three data structure types.
Step 2: Student records are dictionaries for fast search.

Try It (write your answers)

1. Explain: Design a complete program that uses multiple data structures to solve a problem.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Data Structure Project.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Data Structures

2.01 I will give you a list and ask you to write slicing expressions to extract…
2.02 I will give you a list of data and a series of operations to perform
2.03 I will give you a scenario and ask you to design a dictionary that models it
2.04 I will give you a program that accesses dictionary keys without safety checks
2.05 I will give you scenarios and ask whether a tuple or list is more appropriate
2.06 I will give you two lists of data and ask you to find common elements unique…
2.07 I will give you a nested data structure and ask you to access specific deep…
2.08 I will give you an unsorted list and ask you to trace through bubble sort or…
2.09 I will give you a dataset and a search target
2.10 I will give you a real-world scenario and ask you to identify whether it models…
2.11 I will give you a complex problem and ask you to identify all data structures…
2.12 Your project will be evaluated on: correct use of multiple data structures with…

Key terms: List, Index, Slice, Mutability, Method, Append, Pop, Sort, Dictionary, Key

I still need help with:

Chapter Map — Database Fundamentals

- 3.01 What is a Database?
- 3.02 Tables and Records
- 3.03 SQL SELECT
- 3.04 SQL WHERE
- 3.05 SQL ORDER BY
- 3.06 SQL Aggregate Functions
- 3.07 SQL JOIN
- 3.08 Database Design
- 3.09 Primary and Foreign Keys
- 3.10 Database Normalization
- 3.11 CRUD Operations
- 3.12 Database Project

Key terms to know:

Lesson 3.01 What is a Database?

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze a database as an organized collection of structured data stored electronically and justify your choice] [- I can analyze and compare real-world examples of databases used in dAIly life at our school and beyond] [- I can compare file-based storage with database storage explAIning advantages of databases]

Key Words

Data	Facts like numbers and words stored on a computer
Raw Data	Unprocessed facts collected before cleaning or organising properly
Information	Useful meaning produced after processing raw data carefully
Dataset	Collection of related data stored together in one place

Core Idea

A database is an organized collection of structured data stored electronically in a computer system.

Sketch the diagram or concept here.

Worked Example

Step 1: Student records database stores names addresses and parent info used by office staff.
Step 2: Attendance database tracks daily presence used by teachers.

Try It (write your answers)

1. Explain: Analyze a database as an organized collection of structured data stored electronically and justify your choice.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of a Database.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.02 Tables and Records

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand that databases store data in tables with rows called records and columns called fields] [- I can analyze and compare the structure of a table including field names data types and primary keys] [- I can create a simple database table and insert records into it]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

A table is the basic storage unit in a relational database. Each table has columns also called fields or attributes.

Sketch the diagram or concept here.

Worked Example

Step 1: My student table has student_id as the primary key with unique values for each student.
Step 2: The subjects table has subject_id as the primary key.

Try It (write your answers)

1. Explain: Understand that databases store data in tables with rows called records and columns called fields.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Tables and Records.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

_____

Lesson 3.03 SQL SELECT

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can write SELECT statements to retrieve all columns or specific columns from a table] [- I can use SELECT DISTINCT to retrieve unique values and eliminate duplicates] [- I can use column aliases and expressions in SELECT to create computed columns]

Key Words

SELECT	Command to fetch chosen columns from a table for viewing results
Column	Vertical field showing one type of information for all records together
Row	Horizontal record showing all information about one item completely together
Query	Question you ask database to retrieve specific information quickly for answer

Core Idea

The SELECT statement retrieves data from a database table. SELECT star selects all columns. You can specify column names to select only certain columns.

Sketch the diagram or concept here.

Worked Example

Step 1: My first query returns all names and marks.
Step 2: The DISTINCT query shows three unique classes.

Try It (write your answers)

1. Explain: Write SELECT statements to retrieve all columns or specific columns from a table.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of SQL SELECT.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.04 SQL WHERE

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use the WHERE clause to filter records based on specific conditions] [- I can apply comparison operators equals not equals greater than less than in WHERE conditions] [- I can combine multiple conditions using AND and OR operators in WHERE clauses]

Key Words

WHERE	Clause filtering rows to show only those meeting a condition for query
Condition	Rule deciding which rows appear in result like age greater than 15
Filter	Keeping only matching rows and hiding others that do not satisfy condition
Operator	Symbol like equals or greater than used inside condition for comparison

Core Idea

The WHERE clause filters records based on conditions. It follows the SELECT and FROM clauses.

Sketch the diagram or concept here.

Worked Example

Step 1: My queries return specific subsets of the student data.
Step 2: The above 90 query finds high achievers.

Try It (write your answers)

1. Explain: Use the WHERE clause to filter records based on specific conditions.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of SQL WHERE.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.05 SQL ORDER BY

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use ORDER BY to sort query results in ascending or descending order] [- I can sort by multiple columns to handle ties in sorting criteria] [- I can combine ORDER BY with WHERE to filter and sort results simultaneously]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

The ORDER BY clause sorts the result set. By default it sorts in ascending order. Use DESC for descending order and ASC for ascending order explicitly.

Sketch the diagram or concept here.

Worked Example

Step 1: My first query shows Arjun with 98 marks at the top.
Step 2: The alphabetical sort arranges names from A to Z.

Try It (write your answers)

1. Explain: Use ORDER BY to sort query results in ascending or descending order.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of SQL ORDER BY.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.06 SQL Aggregate Functions

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use COUNT to count the number of records matching a condition] [- I can use SUM AVG MAX and MIN to calculate totals averages and find extremes] [- I can use GROUP BY to aggregate data by categories and HAVING to filter groups]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Aggregate functions perform calculations on sets of records. COUNT returns the number of rows. COUNT star counts all rows.

Sketch the diagram or concept here.

Worked Example

Step 1: COUNT with GROUP BY shows Class 8 has 8 students Class 9 has 10 and Class 10 has 7.
Step 2: AVG by class shows Class 10 has the highest average.

Try It (write your answers)

1. Explain: Use COUNT to count the number of records matching a condition.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of SQL Aggregate Functions.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.07 SQL JOIN

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand why JOIN is needed to combine data from multiple related tables] [- I can write INNER JOIN to retrieve records that have matching values in both tables] [- I can use LEFT JOIN to retrieve all records from the left table even without matches]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

JOIN combines rows from two or more tables based on a related column. INNER JOIN returns only rows with matching values in both tables.

Sketch the diagram or concept here.

Worked Example

Step 1: My INNER JOIN shows 20 enrollment records connecting students to subjects.
Step 2: LEFT JOIN shows 25 students with 5 showing NULL for unenrolled students.

Try It (write your answers)

1. Explain: Understand why JOIN is needed to combine data from multiple related tables.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of SQL JOIN.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.08 Database Design

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze and compare the entities and attributes needed for a database from a description] [- I can design a logical database schema with appropriate tables and columns] [- I can create an Entity-Relationship diagram to visualize database structure]

Key Words

Data	Facts like numbers and words stored on a computer
Raw Data	Unprocessed facts collected before cleaning or organising properly
Information	Useful meaning produced after processing raw data carefully
Dataset	Collection of related data stored together in one place

Core Idea

Database design follows a systematic process. First identify the entities which are the main objects to store data about.

Sketch the diagram or concept here.

Worked Example

Step 1: My exam database has four tables.
Step 2: Students store student info.

Try It (write your answers)

1. Explain: Analyze and compare the entities and attributes needed for a database from a description.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Database Design.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.09 Primary and Foreign Keys

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze primary key as a unique identifier for each record in a table and justify your choice] [- I can analyze foreign key as a column that references the primary key of another table and justify your choice] [- I can understand how keys establish relationships between database tables]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

A primary key uniquely identifies each record in a table. It must be unique and cannot be NULL. A table can have only one primary key.

Sketch the diagram or concept here.

Worked Example

Step 1: My teachers table has three teachers with unique teacher_id.
Step 2: My classes table has five classes.

Try It (write your answers)

1. Explain: Analyze primary key as a unique identifier for each record in a table and justify your choice.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Primary and Foreign Keys.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.10 Database Normalization

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand normalization as the process of organizing data to reduce redundancy] [- I can analyze and compare data redundancy and update anomalies in poorly designed tables] [- I can apply basic normalization rules to split a large table into smaller related tables]

Key Words

Data	Facts like numbers and words stored on a computer
Raw Data	Unprocessed facts collected before cleaning or organising properly
Information	Useful meaning produced after processing raw data carefully
Dataset	Collection of related data stored together in one place

Core Idea

Normalization is the process of organizing a database to reduce data redundancy and improve data integrity.

Sketch the diagram or concept here.

Worked Example

Step 1: My normalized design has three tables.
Step 2: Students with student_id and class.

Try It (write your answers)

1. Explain: Understand normalization as the process of organizing data to reduce redundancy.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Database Normalization.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

_____

Lesson 3.11 CRUD Operations

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can implement Create operation using INSERT to add new records to a table] [- I can implement Read operation using SELECT to retrieve data from a table] [- I can implement Update and Delete operations to modify and remove records]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

CRUD represents the four basic database operations. CREATE uses INSERT INTO to add new records. Specify the table columns and values to insert.

Sketch the diagram or concept here.

Worked Example

Step 1: My CRUD application has four functions.
Step 2: add_student inserts a new record.

Try It (write your answers)

1. Explain: Implement Create operation using INSERT to add new records to a table.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of CRUD Operations.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 3.12 Database Project

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can design and implement a complete database for a real-world our school scenario] [- I can apply all SQL concepts including CREATE SELECT WHERE JOIN and aggregates] [- I can document the database design and demonstrate querying capabilities]

Key Words

Data	Facts like numbers and words stored on a computer
Raw Data	Unprocessed facts collected before cleaning or organising properly
Information	Useful meaning produced after processing raw data carefully
Dataset	Collection of related data stored together in one place

Core Idea

A database project follows the complete database lifecycle. Start with requirements analysis to understand what data to store.

Sketch the diagram or concept here.

Worked Example

Step 1: My system has four tables with 25 student records 50 attendance records and 30 result records.
Step 2: My queries find class averages attendance percentages top performers in each subject and students who participate in activities.

Try It (write your answers)

1. Explain: Design and implement a complete database for a real-world our school scenario.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Database Project.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Database Fundamentals

3.01 I will give you a scenario and ask you to design what data a database would…
3.02 I will give you a scenario and ask you to design the table structure including…
3.03 I will give you a database table and ask you to write SELECT statements that…
3.04 I will give you a filtering requirement in plain English and ask you to write…
3.05 I will give you a requirement like find the top 5 students and ask you to write…
3.06 I will give you a data analysis question like which class has the highest…
3.07 I will give you two related tables and ask you to write JOIN queries that…
3.08 I will give you a real-world scenario and ask you to design a complete database
3.09 I will give you a scenario and ask you to identify the primary key for each…
3.10 I will give you a table with redundant data and ask you to normalize it
3.11 I will give you a database scenario and ask you to write CRUD operations for…
3.12 Your project will be evaluated on: database design quality with appropriate…

Key terms: Data, Raw Data, Information, Dataset, Concept, Example, Practice, Apply, SELECT, Column

I still need help with:

Chapter Map — Web Technologies

- 4.01 How the Web Works
- 4.02 HTML Basics
- 4.03 HTML Elements
- 4.04 HTML Forms
- 4.05 CSS Introduction
- 4.06 CSS Selectors
- 4.07 CSS Layout
- 4.08 Responsive Design
- 4.09 JavaScript Introduction
- 4.10 DOM Manipulation
- 4.11 Web Security
- 4.12 Portfolio Website

Key terms to know:

Lesson 4.01 How the Web Works

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn the client-server model and how browsers request web pages] [- I can understand the role of URLs IP addresses and DNS in web navigation] [- I can describe the HTTP request-response cycle that powers web communication]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

The web operates on a client-server model. The client is your web browser like Chrome or Firefox. The server is a computer that stores websites.

Sketch the diagram or concept here.

Worked Example

Step 1: My diagram shows six steps from typing the URL to seeing the page.
Step 2: DNS is step one translating the name.

Try It (write your answers)

1. Explain: Explain the client-server model and how browsers request web pages.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of How the Web Works.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.02 HTML Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can write a basic HTML document with proper structure including head and body] [- I can use heading paragraph and line break tags to structure content] [- I can open and view HTML files in a web browser to see the rendered output]

Key Words

HTML	Language using tags to build structure of web pages for browsers
Tag	Word inside angle brackets marking start and end of element like <p>
Element	Complete part made of opening tag content and closing tag together
Attribute	Extra information inside tag like colour or link address for control

Core Idea

HTML uses tags to mark up content. Tags come in pairs with an opening tag and a closing tag. The closing tag has a forward slash.

Sketch the diagram or concept here.

Worked Example

Step 1: My HTML page has the school name as the main heading.
Step 2: About Our School section describes the school in two paragraphs.

Try It (write your answers)

1. Explain: Write a basic HTML document with proper structure including head and body.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of HTML Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.03 HTML Elements

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use list tags to create ordered and unordered lists on web pages] [- I can add images to HTML pages using the img tag with src and alt attributes] [- I can create hyperlinks using anchor tags to connect web pages together]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

HTML elements extend basic text with rich content. The ul element creates unordered lists with bullet points.

Sketch the diagram or concept here.

Worked Example

Step 1: My page has a bullet list of facilities including library lab and playground.
Step 2: My numbered list shows the daily schedule.

Try It (write your answers)

1. Explain: Use list tags to create ordered and unordered lists on web pages.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of HTML Elements.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.04 HTML Forms

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can create HTML forms with input fields for text emAIl and password data] [- I can use form controls like buttons dropdowns and checkboxes for user interaction] [- I can understand the role of forms in collecting user input for web applications]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

HTML forms collect user input. The form element is the container for all form controls.

Sketch the diagram or concept here.

Worked Example

Step 1: My admission form collects all necessary information.
Step 2: Name and parent name use text inputs.

Try It (write your answers)

1. Explain: Create HTML forms with input fields for text email and password data.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of HTML Forms.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.05 CSS Introduction

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand that CSS controls the visual presentation of HTML elements] [- I can apply inline internal and external CSS to style web pages] [- I can use CSS properties to change colors fonts and backgrounds]

Key Words

CSS	Language styling HTML by setting colours fonts and layout beautifully together
Selector	Pattern choosing which HTML elements to style together as group
Property	Aspect like colour or margin you want to change for selected elements
Rule	Complete instruction pairing selector with property and value for styling

Core Idea

CSS controls the visual presentation of HTML elements. Inline CSS uses the style attribute on individual elements.

Sketch the diagram or concept here.

Worked Example

Step 1: My CSS file sets the background to light blue.
Step 2: Headings are dark green and centered.

Try It (write your answers)

1. Explain: Understand that CSS controls the visual presentation of HTML elements.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of CSS Introduction.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.06 CSS Selectors

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use element class and ID selectors to target specific HTML elements] [- I can understand selector specificity and how it determines which styles apply] [- I can apply pseudo-class selectors to style elements based on user interaction]

Key Words

CSS	Language styling HTML by setting colours fonts and layout beautifully together
Selector	Pattern choosing which HTML elements to style together as group
Property	Aspect like colour or margin you want to change for selected elements
Rule	Complete instruction pairing selector with property and value for styling

Core Idea

CSS selectors determine which elements receive styles. Element targets all of a type. Class uses a dot prefix. ID uses a hash prefix.

Sketch the diagram or concept here.

Worked Example

Step 1: My page uses element selector for paragraphs the special class for highlighted content the ID for the heading and hover effects on links.

Try It (write your answers)

1. Explain: Use element class and ID selectors to target specific HTML elements.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of CSS Selectors.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.07 CSS Layout

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can use the box model concept including margin border padding and content area] [- I can apply dISPlay property values like block inline and flex to control layout] [- I can create multi-column layouts using CSS flexbox for modern web design]

Key Words

CSS	Language styling HTML by setting colours fonts and layout beautifully together
Selector	Pattern choosing which HTML elements to style together as group
Property	Aspect like colour or margin you want to change for selected elements
Rule	Complete instruction pairing selector with property and value for styling

Core Idea

The box model has content padding border and margin. Content is actual text. Padding is space inside the border. Margin is space outside.

Sketch the diagram or concept here.

Worked Example

Step 1: My layout has a green header spanning full width.
Step 2: Sidebar takes 25 percent and content takes 75 percent.

Try It (write your answers)

1. Explain: Use the box model concept including margin border padding and content area.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of CSS Layout.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.08 Responsive Design

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand why web pages need to work on different screen sizes] [- I can use media queries to apply different CSS rules based on screen width] [- I can apply the viewport meta tag and relative units for mobile-friendly design]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Responsive design ensures pages work on all devices. Viewport meta tag tells the browser to use actual width.

Sketch the diagram or concept here.

Worked Example

Step 1: My website now has the viewport tag.
Step 2: Media query stacks sidebar below content on small screens.

Try It (write your answers)

1. Explain: Understand why web pages need to work on different screen sizes.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Responsive Design.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.09 JavaScript Introduction

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand that JavaScript adds interactivity and behavior to web pages] [- I can write JavaScript variables loops and conditionals for client-side logic] [- I can use the alert function and console to dISPlay information and debug code]

Key Words

JavaScript	Language adding interactivity like clicks and animations to web pages dynamically
Script	List of instructions running inside browser to make page interactive for users
Event	Action like click or key press that triggers JavaScript code to run
DOM	Tree structure representing webpage that JavaScript can change live instantly

Core Idea

JavaScript is a programming language that runs in browsers. Write in script tags or external js files. Variables store data using let and const.

Sketch the diagram or concept here.

Worked Example

Step 1: My program stores the name in a variable.
Step 2: If-else checks for empty name.

Try It (write your answers)

1. Explain: Understand that JavaScript adds interactivity and behavior to web pages.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of JavaScript Introduction.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.10 DOM Manipulation

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand the Document Object Model as a tree of HTML elements] [- I can use getElementById and querySelector to select HTML elements] [- I can modify element content styles and attributes dynamically]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

The DOM is a tree of HTML elements the browser creates. getElementById finds by ID. querySelector finds by CSS selector. textContent changes text.

Sketch the diagram or concept here.

Worked Example

Step 1: My button changes background from white to blue on click.
Step 2: Each click cycles through colors.

Try It (write your answers)

1. Explain: Understand the Document Object Model as a tree of HTML elements.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of DOM Manipulation.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.11 Web Security

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze and compare common web threats including phishing and cross-site scripting] [- I can understand HTTPS encryption and why secure connections protect data] [- I can apply safe browsing habits to protect personal information online]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Web security threats include phishing which tricks users into revealing data and malware which damages devices. HTTPS encrypts communication.

Sketch the diagram or concept here.

Worked Example

Step 1: My poster includes password tips recognizing phishing by checking sender addresses and looking for HTTPS lock icon.

Try It (write your answers)

1. Explain: Analyze and compare common web threats including phishing and cross-site scripting.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Web Security.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 4.12 Portfolio Website

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can combine HTML CSS and JavaScript to build a complete personal portfolio website] [- I can apply responsive design principles to ensure the site works on all devices] [- I can present and document the website with design decisions and technical choices]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

A portfolio website demonstrates all web technologies working together. HTML creates structure with semantic elements.

Sketch the diagram or concept here.

Worked Example

Step 1: My portfolio has home about skills and contact sections.
Step 2: CSS flexbox creates layout.

Try It (write your answers)

1. Explain: Combine HTML CSS and JavaScript to build a complete personal portfolio website.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Portfolio Website.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Web Technologies

4.01 I will ask you to explain the complete process of visiting a website from…
4.02 I will give you a content description and ask you to write the correct HTML…
4.03 I will give you a content description and ask you to write the correct HTML…
4.04 I will give you a form description and ask you to write the HTML code
4.05 I will give you an HTML page and ask you to write CSS to achieve a specific…
4.06 Write CSS using correct selectors to target specific elements without affecting…
4.07 Write CSS to achieve a described page layout using box model and flexbox
4.08 Make a desktop layout responsive by adding viewport tag media queries and…
4.09 Write JavaScript using variables conditionals and loops to solve a problem
4.10 Write JavaScript that selects HTML elements and modifies their content or style
4.11 Analyze and compare security threats in given scenarios and recommend…
4.12 Your project will be evaluated on proper HTML structure correct CSS styling…

Key terms: Concept, Example, Practice, Apply, HTML, Tag, Element, Attribute, CSS, Selector

I still need help with:

Chapter Map — Algorithms and Computational Thinking

- 5.01 What is an Algorithm?
- 5.02 Flowcharts
- 5.03 Pseudocode
- 5.04 Sequential Algorithms
- 5.05 Selection Algorithms
- 5.06 Iteration Algorithms
- 5.07 Big-O Notation
- 5.08 Divide and Conquer
- 5.09 Greedy Algorithms
- 5.10 Algorithm Efficiency
- 5.11 Problem Decomposition
- 5.12 Algorithm Design Project

Key terms to know:

Lesson 5.01 What is an Algorithm?

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze an algorithm as a step-by-step procedure to solve a problem and justify your choice] [- I can analyze and compare algorithms in everyday activities like cooking and directions] [- I can write simple algorithms for real-world tasks using clear sequential steps]

Key Words

Algorithm	Step-by-step instructions solving a problem clearly and completely in order
Flowchart	Diagram using boxes and arrows showing algorithm steps visually for understanding
Pseudocode	Simple English-like description of algorithm before writing real code for planning
Efficiency	How quickly algorithm finishes and how little memory it uses together

Core Idea

An algorithm is a finite set of unambiguous steps that solve a specific problem. Key properties are finite unambiguous and effective.

Sketch the diagram or concept here.

Worked Example

Step 1: My shoelace algorithm has 7 steps.
Step 2: My bag packing checks the timetable first.

Try It (write your answers)

1. Explain: Analyze an algorithm as a step-by-step procedure to solve a problem and justify your choice.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of an Algorithm.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.02 Flowcharts

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can draw flowcharts using standard symbols for start end process and decision] [- I can convert a written algorithm into a visual flowchart representation] [- I can read and interpret flowcharts to understand program logic]

Key Words

Chart	Visual picture showing numbers as bars lines or pies for easy comparison
Bar Chart	Rectangles whose height shows value of each category clearly for viewer
Pie Chart	Circle divided into slices showing share of each part
Legend	Key matching colours to names so chart is understandable quickly

Core Idea

Flowcharts use standard symbols. Oval represents start and end. Rectangle represents process. Diamond represents decision with two exits.

Sketch the diagram or concept here.

Worked Example

Step 1: My flowchart starts with input three numbers.
Step 2: First diamond compares A and B.

Try It (write your answers)

1. Explain: Draw flowcharts using standard symbols for start end process and decision.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Flowcharts.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.03 Pseudocode

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can write pseudocode to express algorithms in a structured but readable format] [- I can use pseudocode keywords like BEGIN END IF THEN ELSE and WHILE DO] [- I can convert between pseudocode and flowchart representations]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Pseudocode is an informal high-level description of an algorithm. It uses keywords like BEGIN END IF THEN ELSE WHILE DO FOR and RETURN.

Sketch the diagram or concept here.

Worked Example

Step 1: My average algorithm computes sum then divides by 3.
Step 2: My even-odd uses MOD operator.

Try It (write your answers)

1. Explain: Write pseudocode to express algorithms in a structured but readable format.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Pseudocode.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.04 Sequential Algorithms

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze and compare sequential algorithms where steps follow one after another in order] [- I can write sequential algorithms for problems with a definite beginning and end] [- I can trace through sequential algorithms step by step to verify correctness]

Key Words

Algorithm	Step-by-step instructions solving a problem clearly and completely in order
Flowchart	Diagram using boxes and arrows showing algorithm steps visually for understanding
Pseudocode	Simple English-like description of algorithm before writing real code for planning
Efficiency	How quickly algorithm finishes and how little memory it uses together

Core Idea

Sequential algorithms have steps that execute in order from first to last. There are no decisions or repetitions.

Sketch the diagram or concept here.

Worked Example

Step 1: My area algorithm multiplies length times width.
Step 2: My temperature uses F equals C times 9/5 plus 32.

Try It (write your answers)

1. Explain: Analyze and compare sequential algorithms where steps follow one after another in order.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Sequential Algorithms.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.05 Selection Algorithms

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze and compare selection algorithms that make decisions using IF THEN ELSE] [- I can write algorithms with single and nested selection structures] [- I can trace selection algorithms to determine which path is taken for given inputs]

Key Words

Algorithm	Step-by-step instructions solving a problem clearly and completely in order
Flowchart	Diagram using boxes and arrows showing algorithm steps visually for understanding
Pseudocode	Simple English-like description of algorithm before writing real code for planning
Efficiency	How quickly algorithm finishes and how little memory it uses together

Core Idea

Selection algorithms use conditions to choose between paths. IF condition THEN action ELSE alternative. Nested selection adds more conditions.

Sketch the diagram or concept here.

Worked Example

Step 1: My grading uses ranges.
Step 2: Leap year checks divisibility by 4 and not by 100 or by 400.

Try It (write your answers)

1. Explain: Analyze and compare selection algorithms that make decisions using IF THEN ELSE.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Selection Algorithms.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.06 Iteration Algorithms

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze and compare iteration as repetition of steps using loops] [- I can write algorithms with FOR loops for known repetitions and WHILE loops for unknown] [- I can trace iteration algorithms to count loop executions and track variable changes]

Key Words

Algorithm	Step-by-step instructions solving a problem clearly and completely in order
Flowchart	Diagram using boxes and arrows showing algorithm steps visually for understanding
Pseudocode	Simple English-like description of algorithm before writing real code for planning
Efficiency	How quickly algorithm finishes and how little memory it uses together

Core Idea

FOR loops repeat a known number of times. WHILE loops repeat until a condition becomes false.

Sketch the diagram or concept here.

Worked Example

Step 1: My multiplication table uses FOR loop from 1 to 10.
Step 2: My sum uses WHILE adding each number.

Try It (write your answers)

1. Explain: Analyze and compare iteration as repetition of steps using loops.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Iteration Algorithms.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.07 Big-O Notation

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand Big-O notation as a way to describe algorithm efficiency] [- I can analyze and compare common time complexities: O(1) O(n) O(n squared) and O(log n)] [- I can compare algorithms based on their growth rate and efficiency for large inputs]

Key Words

Big-O	Notation describing how runtime grows when input size increases dramatically for comparison
Complexity	Amount of time or memory algorithm needs as data grows larger
Linear	Time growing directly with input size like checking each item once
Constant	Time staying same no matter how large input becomes for operation

Core Idea

Big-O describes the worst-case growth rate. O(1) constant means same time regardless of input. O(n) linear means time grows with input.

Sketch the diagram or concept here.

Worked Example

Step 1: Checking every element is O(n).
Step 2: Binary search is O(log n).

Try It (write your answers)

1. Explain: Understand Big-O notation as a way to describe algorithm efficiency.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Big-O Notation.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.08 Divide and Conquer

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand divide and conquer as breaking a problem into smaller subproblems] [- I can analyze and compare how merge sort and binary search use divide and conquer strategy] [- I can compare divide and conquer approaches with brute force approaches]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Divide and conquer algorithms break a problem into subproblems solve each recursively and combine results.

Sketch the diagram or concept here.

Worked Example

Step 1: Split 8 into two 4s.
Step 2: Find largest in each.

Try It (write your answers)

1. Explain: Understand divide and conquer as breaking a problem into smaller subproblems.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Divide and Conquer.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.09 Greedy Algorithms

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand greedy algorithms that make the best local choice at each step] [- I can apply greedy strategy to problems like making change and activity selection] [- I can analyze and compare when greedy algorithms work well and when they fAIl]

Key Words

Algorithm	Step-by-step instructions solving a problem clearly and completely in order
Flowchart	Diagram using boxes and arrows showing algorithm steps visually for understanding
Pseudocode	Simple English-like description of algorithm before writing real code for planning
Efficiency	How quickly algorithm finishes and how little memory it uses together

Core Idea

Greedy algorithms make the locally optimal choice at each step hoping for a globally optimal result.

Sketch the diagram or concept here.

Worked Example

Step 1: For activities I sort by end time and pick non-overlapping ones.
Step 2: For change 63 = 50 + 10 + 1 + 1 + 1 = 5 coins.

Try It (write your answers)

1. Explain: Understand greedy algorithms that make the best local choice at each step.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Greedy Algorithms.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.10 Algorithm Efficiency

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can compare the efficiency of different algorithms for the same problem] [- I can understand that efficiency affects how programs handle large datasets] [- I can choose the most efficient algorithm based on problem requirements]

Key Words

Algorithm	Step-by-step instructions solving a problem clearly and completely in order
Flowchart	Diagram using boxes and arrows showing algorithm steps visually for understanding
Pseudocode	Simple English-like description of algorithm before writing real code for planning
Efficiency	How quickly algorithm finishes and how little memory it uses together

Core Idea

Efficiency is measured by time complexity and space complexity. Time complexity measures how operations grow. Space complexity measures memory usage.

Sketch the diagram or concept here.

Worked Example

Step 1: Check every pair is simple but slow.
Step 2: Sort is faster.

Try It (write your answers)

1. Explain: Compare the efficiency of different algorithms for the same problem.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Algorithm Efficiency.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.11 Problem Decomposition

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can break complex problems into smaller manageable subproblems] [- I can analyze and compare how smaller solutions combine to solve the original problem] [- I can apply decomposition to real-world our school scenarios]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Decomposition is breaking a complex problem into simpler subproblems. Each subproblem can be solved independently.

Sketch the diagram or concept here.

Worked Example

Step 1: Library system needs catalog, borrowing, and returns.
Step 2: Event needs venue, schedule, and invitations.

Try It (write your answers)

1. Explain: Break complex problems into smaller manageable subproblems.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Problem Decomposition.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 5.12 Algorithm Design Project

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can apply algorithmic thinking to design a complete solution for a real problem] [- I can use flowcharts pseudocode and efficiency analysis in the design process] [- I can document the design process and justify algorithmic choices]

Key Words

Algorithm	Step-by-step instructions solving a problem clearly and completely in order
Flowchart	Diagram using boxes and arrows showing algorithm steps visually for understanding
Pseudocode	Simple English-like description of algorithm before writing real code for planning
Efficiency	How quickly algorithm finishes and how little memory it uses together

Core Idea

A complete algorithm design project follows these steps: understand the problem, decompose into subproblems, design algorithms for each, represent wit

Sketch the diagram or concept here.

Worked Example

Step 1: My attendance system decomposes into: mark attendance view records generate report.
Step 2: Each has efficient algorithms.

Try It (write your answers)

1. Explain: Apply algorithmic thinking to design a complete solution for a real problem.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Algorithm Design Project.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Algorithms and Computational Thinking

5.01 Write a complete algorithm for a given task
5.02 Draw a flowchart for a given algorithm and describe what a given flowchart does
5.03 Convert a given flowchart to pseudocode and verify both represent the same logic
5.04 Given a problem write a sequential algorithm and trace it with test values
5.05 Write a selection algorithm and trace it with different inputs to verify all…
5.06 Write an iteration algorithm and trace it counting each loop execution
5.07 Given an algorithm determine its Big-O notation and explain why
5.08 Analyze and compare whether a given algorithm uses divide and conquer and…
5.09 Determine if a greedy approach works for a given problem and justify your answer
5.10 For a given problem compare different algorithm approaches and recommend the…
5.11 Given a complex problem decompose it into subproblems and describe how solving…
5.12 Present your design explaining each algorithm choice and efficiency analysis

Key terms: Algorithm, Flowchart, Pseudocode, Efficiency, Chart, Bar Chart, Pie Chart, Legend, Concept, Example

I still need help with:

Chapter Map — Emerging Technologies

- 6.01 What is AI?
- 6.02 Machine Learning Basics
- 6.03 Neural Networks
- 6.04 IoT Introduction
- 6.05 Smart Devices
- 6.06 Cloud Computing Advanced
- 6.07 Blockchain Basics
- 6.08 Quantum Computing Intro
- 6.09 Ethics in Technology
- 6.10 Digital Divide
- 6.11 Future Careers
- 6.12 Technology Impact Essay

Key terms to know:

Lesson 6.01 What is AI?

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze artificial intelligence as machines that can perform tasks requiring human intelligence and justify your choice] [- I can analyze and compare examples of AI in dAIly life including voice assistants and recommendation systems] [- I can distinguish between narrow AI and general AI capabilities]

Key Words

AI	Machines mimicking human thinking to solve problems without explicit instructions always
Model	Trained system that makes predictions based on data it learned from examples
Data	Examples used to teach AI system patterns for making decisions later
Ethics	Rules ensuring fair and safe use of AI for everyone

Core Idea

Artificial intelligence is machines performing tasks that normally require human intelligence.

Sketch the diagram or concept here.

Worked Example

Step 1: I researched AI in Indian agriculture.
Step 2: It analyzes soil and weather data to advise farmers.

Try It (write your answers)

1. Explain: Analyze artificial intelligence as machines that can perform tasks requiring human intelligence and justify your choice.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of AI.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.02 Machine Learning Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can explAIn machine learning as AI that learns from data without being explicitly programmed] [- I can analyze and compare three types of machine learning: supervised unservised and reinforcement] [- I can understand how trAIning data is used to build machine learning models]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Machine learning is AI that improves through experience. Supervised learning uses labeled training data with correct answers.

Sketch the diagram or concept here.

Worked Example

Step 1: This is supervised learning.
Step 2: I would use past marks attendance and homework data.

Try It (write your answers)

1. Explain: Explain machine learning as AI that learns from data without being explicitly programmed.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Machine Learning Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

Lesson 6.03 Neural Networks

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand neural networks as computing systems inspired by the human brAIn] [- I can explAIn how neural networks process information through layers of connected nodes] [- I can analyze and compare applications of neural networks in image recognition and language processing]

Key Words

Neural Network	System inspired by brain with layers of connected nodes learning patterns together
Neuron	Single unit receiving inputs and passing result to next layer for processing
Layer	Group of neurons processing together at same stage before passing onward
Training	Adjusting connections by showing many examples until accuracy improves gradually step

Core Idea

Neural networks are layers of interconnected nodes inspired by biological neurons. Input layer receives data. Hidden layers process it.

Sketch the diagram or concept here.

Worked Example

Step 1: Handwriting recognition uses input nodes for pixel values.
Step 2: Hidden layers detect features like curves and lines.

Try It (write your answers)

1. Explain: Understand neural networks as computing systems inspired by the human brain.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Neural Networks.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.04 IoT Introduction

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze Internet of Things as everyday objects connected to the internet to collect and share data and justify your choice] [- I can analyze and compare IoT devices in smart homes schools and cities] [- I can understand how IoT sensors collect data and send it for processing]

Key Words

IoT	Everyday objects like bulbs connected to internet to be controlled remotely easily
Sensor	Device detecting temperature light or motion and sending data to system
Actuator	Device that moves or switches something after receiving command from system
Smart Device	Object that senses acts and connects to internet automatically without human help

Core Idea

IoT connects physical objects to the internet. Devices have sensors to collect data processors to analyze it and network connections to share it.

Sketch the diagram or concept here.

Worked Example

Step 1: My smart school has attendance sensors at the door temperature sensors in classrooms and energy sensors on lights.
Step 2: All data goes to a central dashboard.

Try It (write your answers)

1. Explain: Analyze Internet of Things as everyday objects connected to the internet to collect and share data and justify your choice.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of IoT Introduction.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.05 Smart Devices

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand how smart devices use sensors processors and connectivity to automate tasks] [- I can compare smart devices with traditional devices and identify the advantages] [- I can analyze the impact of smart devices on dAIly life at our school and in homes]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Smart devices have four components: sensors to collect data processors to analyze it connectivity to share data and actuators to take action.

Sketch the diagram or concept here.

Worked Example

Step 1: Smart attendance uses facial recognition benefits include accuracy concerns include privacy.
Step 2: Smart classroom uses interactive boards benefits include engagement.

Try It (write your answers)

1. Explain: Understand how smart devices use sensors processors and connectivity to automate tasks.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Smart Devices.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.06 Cloud Computing Advanced

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand cloud computing as storing and accessing data and programs over the internet] [- I can compare IaaS PaaS and SaaS cloud service models with real-world examples] [- I can analyze the benefits of cloud computing for schools including cost and scalability]

Key Words

Cloud	Servers on internet that store your files safely always
Server	Powerful computer storing many files for many users together
Upload	Sending your file from device to cloud storage
Download	Bringing a file from cloud to your own device

Core Idea

Cloud computing delivers computing services over the internet. IaaS provides virtual machines and storage. PaaS provides development platforms.

Sketch the diagram or concept here.

Worked Example

Step 1: Google Classroom is SaaS for learning.
Step 2: Google Drive is SaaS for storage.

Try It (write your answers)

1. Explain: Understand cloud computing as storing and accessing data and programs over the internet.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Cloud Computing Advanced.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

_____

I still need help with:

Lesson 6.07 Blockchain Basics

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand blockchAIn as a distributed digital ledger that records transactions securely] [- I can explAIn how blockchAIn ensures data cannot be altered once recorded] [- I can analyze and compare applications of blockchAIn beyond cryptocurrency including supply chAIn and voting]

Key Words

AI	Machines mimicking human thinking to solve problems without explicit instructions always
Model	Trained system that makes predictions based on data it learned from examples
Data	Examples used to teach AI system patterns for making decisions later
Ethics	Rules ensuring fair and safe use of AI for everyone

Core Idea

Blockchain is a distributed ledger across many computers. Each block contains transactions and a hash linking to the previous block.

Sketch the diagram or concept here.

Worked Example

Step 1: Supply chain tracks products from farm to table.
Step 2: Medical records ensure data integrity.

Try It (write your answers)

1. Explain: Understand blockchain as a distributed digital ledger that records transactions securely.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Blockchain Basics.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.08 Quantum Computing Intro

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can understand quantum computing as a new type of computing using quantum mechanical phenomena] [- I can explAIn the concepts of qubits superposition and entanglement in simple terms] [- I can compare quantum computing potential with classical computing limitations]

Key Words

Quantum	Computing using tiny particles that hold many states at once
Qubit	Unit holding zero one or both together enabling powerful calculations beyond normal
Superposition	Being in multiple states simultaneously until measured for quantum advantage greatly
Entanglement	Link where two qubits affect each other instantly even when far apart

Core Idea

Quantum computing uses qubits which exploit superposition being in multiple states at once and entanglement where qubits are connected.

Sketch the diagram or concept here.

Worked Example

Step 1: Drug discovery requires simulating molecular interactions.
Step 2: Classical computers cannot handle the complexity.

Try It (write your answers)

1. Explain: Understand quantum computing as a new type of computing using quantum mechanical phenomena.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Quantum Computing Intro.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.09 Ethics in Technology

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze and compare ethical issues in technology including privacy bias and fAIrness] [- I can understand the importance of responsible technology development and use] [- I can analyze real-world scenarios involving technology ethics and propose solutions]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Technology ethics covers privacy fair access algorithmic bias intellectual property and environmental impact.

Sketch the diagram or concept here.

Worked Example

Step 1: Benefits include accuracy and automation.
Step 2: Risks include privacy violation and data security.

Try It (write your answers)

1. Explain: Analyze and compare ethical issues in technology including privacy bias and fairness.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Ethics in Technology.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.10 Digital Divide

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze digital divide as the gap between those with and without access to technology and justify your choice] [- I can analyze and compare causes and consequences of the digital divide in India and globally] [- I can propose solutions to bridge the digital divide and ensure equitable access]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

The digital divide has three aspects: access to devices availability of internet and digital literacy.

Sketch the diagram or concept here.

Worked Example

Step 1: Solutions include shared computer labs mobile device lending libraries and digital literacy programs.

Try It (write your answers)

1. Explain: Analyze digital divide as the gap between those with and without access to technology and justify your choice.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Digital Divide.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.11 Future Careers

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can analyze and compare careers emerging from technology including AI specialist data scientist and cybersecurity analyst] [- I can understand how technology is changing traditional careers and creating new ones] [- I can explore the skills needed for technology careers and plan learning paths]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

Technology is transforming all careers. Traditional jobs now require digital skills.

Sketch the diagram or concept here.

Worked Example

Step 1: My plan includes learning Python for AI studying data analysis for data science and understanding networks for cybersecurity.

Try It (write your answers)

1. Explain: Analyze and compare careers emerging from technology including AI specialist data scientist and cybersecurity analyst.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Future Careers.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Lesson 6.12 Technology Impact Essay

Name: _____ Class: _____ Date: _____ / _____ / _____

I Will Be Able To...

[- I can write a structured essay analyzing the impact of technology on society] [- I can support arguments with evidence from the technology topics studied in this chapter] [- I can present balanced perspectives on both benefits and risks of technology]

Key Words

Concept	Main idea that helps you understand the lesson topic clearly and correctly
Example	Specific case showing how the concept works in real life situations
Practice	Repeating task many times to improve skill and remember better always
Apply	Using what you learned to solve new problem or daily task usefully

Core Idea

A technology impact essay has a clear thesis statement evidence-based body paragraphs and a thoughtful conclusion.

Sketch the diagram or concept here.

Worked Example

Step 1: My essay argues that technology brings great benefits but requires responsible use.
Step 2: I covered AI IoT ethics and digital divide with specific examples.

Try It (write your answers)

1. Explain: Write a structured essay analyzing the impact of technology on society.

Answer: Use the Core Idea and Worked Example to help.

2. Give one real-life example of Technology Impact Essay.

Answer: Any correct example from your daily life.

Check Yourself

1. What is the most important thing you learned today?

Answer: Your own answer.

2. Give one example of how this topic is used in real life.

Answer: Any correct real-life example.

Today I Learned

Write one thing you learned today:

I still need help with:

Quick Revision — Emerging Technologies

6.01 For each AI example identify the human intelligence it mimics and discuss…
6.02 For a given scenario identify whether it uses supervised unsupervised or…
6.03 Explain how a neural network processes information from input to output for a…
6.04 For a given scenario identify the IoT devices sensors data collected and…
6.05 Evaluate the benefits and drawbacks of adopting smart devices in a school…
6.06 Compare cloud computing with local computing and recommend when each is…
6.07 Explain how blockchain achieves trust and security without a central authority
6.08 Explain the potential advantages and current limitations of quantum computing
6.09 Propose ethical guidelines for technology use at our school considering privacy…
6.10 Evaluate the impact of the digital divide on education and propose policy…
6.11 Compare a traditional career path with a technology career path and identify…
6.12 Evaluate peer essays for thesis clarity evidence use and balanced perspective

Key terms: AI, Model, Data, Ethics, Concept, Example, Practice, Apply, Neural Network, Neuron

I still need help with:

Glossary

AI	Machines mimicking human thinking to solve problems without explicit instructions always
Actuator	Device that moves or switches something after receiving command from system
Algorithm	Step-by-step instructions solving a problem clearly and completely in order
Append	Adding item to end of list without changing other elements quickly
Apply	Using what you learned to solve new problem or daily task usefully
Argument	Actual value you pass to function when you call it for execution
Array	Ordered list storing many values of same type under one name
Attribute	Extra information inside tag like colour or link address for control
Bar Chart	Rectangles whose height shows value of each category clearly for viewer
Base Case	Simple situation where recursion stops and returns answer directly without further calls
Big-O	Notation describing how runtime grows when input size increases dramatically for comparison
Binary Search	Repeatedly halving sorted list to find item much faster than linear
Bubble Sort	Repeatedly swapping neighbours that are out of order until list sorted completely
Bug	Mistake in code that makes program behave incorrectly or stop
CSS	Language styling HTML by setting colours fonts and layout beautifully together
Call	Telling program to run the function by writing its name with brackets
Caller	Code that calls function and waits to receive returned result back
Chart	Visual picture showing numbers as bars lines or pies for easy comparison
Cloud	Servers on internet that store your files safely always
Code	Written instructions that computer can understand and follow exactly
Cohesion	How well parts inside one module work together for single purpose
Column	Vertical field showing one type of information for all records together
Comparison	Checking which of two items is smaller to decide swapping during sort
Complexity	Measure of how many steps sorting needs as list grows bigger gradually
Concatenation	Joining two strings together end to end to make one longer string
Concept	Main idea that helps you understand the lesson topic clearly and correctly
Condition	Rule deciding which rows appear in result like age greater than 15
Constant	Time staying same no matter how large input becomes for operation
DOM	Tree structure representing webpage that JavaScript can change live instantly
DRY	Do Not Repeat Yourself principle encouraging reuse of code efficiently
Data	Facts like numbers and words stored on a computer
Dataset	Collection of related data stored together in one place
Definition	Writing function code once so it can be called later anywhere
Delete	Removing element from array and shifting others to fill gap
Dictionary	Collection storing pairs of key and value for quick lookup by name
Download	Bringing a file from cloud to your own device
Efficiency	How quickly algorithm finishes and how little memory it uses together
Element	Single value stored at one position inside array for access later
Entanglement	Link where two qubits affect each other instantly even when far apart
Ethics	Rules ensuring fair and safe use of AI for everyone
Event	Action like click or key press that triggers JavaScript code to run
Example	Specific case showing how the concept works in real life situations
Except	Block running alternative code when try block fails with error gracefully
Exception	Error happening during program run that would stop execution if unhandled
File	Collection of data stored on disk with a name like notes.txt
Filter	Keeping only matching rows and hiding others that do not satisfy condition
Finally	Block always running whether error happened or not for cleanup

Flowchart	Diagram using boxes and arrows showing algorithm steps visually for understanding
Function	Reusable block of code that performs one specific task perfectly each time
HTML	Language using tags to build structure of web pages for browsers
Immutable	Cannot be altered after creation protecting data from accidental change completely
Import	Bringing module code into current program to use its functions
Index	Number showing position of element starting from zero in array quickly
Information	Useful meaning produced after processing raw data carefully
Inner Loop	Second loop running fully each time outer takes one step forward
Input	Data given to function so it works with different values
Insert	Adding new element at specific position inside existing array carefully
Instruction	Single command telling computer to do one specific action
Intersection	Keeping only elements present in both sets together as common part
IoT	Everyday objects like bulbs connected to internet to be controlled remotely easily
Iterate	Visiting each key-value pair one by one to process them all sequentially
Iteration	Single execution of loop body before checking condition again for next
JavaScript	Language adding interactivity like clicks and animations to web pages dynamically
Key	Unique name used to find its matching value inside dictionary quickly
Layer	Group of neurons processing together at same stage before passing onward
Legend	Key matching colours to names so chart is understandable quickly
Length	Number of characters including spaces inside a string counted together
Linear	Time growing directly with input size like checking each item once
Linear Search	Checking each item one by one from start until found sequentially
List	Ordered collection holding many values that can change during program run
Lookup	Finding value by giving its key to dictionary quickly without scanning
Method	Function belonging to list called with dot like list.append for action
Model	Trained system that makes predictions based on data it learned from examples
Module	File containing related functions that can be reused in many programs
Mutability	Ability to change list contents after creation without making new list
Nested Loop	Loop placed inside another loop running inner many times per outer
Neural Network	System inspired by brain with layers of connected nodes learning patterns together
Neuron	Single unit receiving inputs and passing result to next layer for processing
None	Special value meaning function returned nothing useful for caller to use
Operator	Symbol like equals or greater than used inside condition for comparison
Outer Loop	First loop controlling how many times inner repeats completely each cycle
Packing	Putting many values together into one tuple in single step conveniently
Pair	Single key together with its value as one entry inside dictionary
Parameter	Variable listed in function definition receiving value when called correctly for flexibility
Path	Address showing where file is stored like folder inside folder
Pie Chart	Circle divided into slices showing share of each part
Pop	Removing and returning last item from list for use elsewhere immediately
Practice	Repeating task many times to improve skill and remember better always
Programming	Writing instructions so computer performs tasks step by step
Property	Aspect like colour or margin you want to change for selected elements
Pseudocode	Simple English-like description of algorithm before writing real code for planning
Push	Adding new item to top of stack for storage immediately quickly
Quantum	Computing using tiny particles that hold many states at once
Qubit	Unit holding zero one or both together enabling powerful calculations beyond normal
Query	Question you ask database to retrieve specific information quickly for answer

Queue	Line where first added item is removed first like school queue
Raw Data	Unprocessed facts collected before cleaning or organising properly
Read	Getting data from file into program for processing or display on screen
Recursion	Function calling itself to solve smaller version of same problem cleverly
Recursive Call	Function calling itself with smaller input to move toward base case gradually
Return	Sending result back from function to the place that called it
Reuse	Using same code in many places without writing again for efficiency
Row	Horizontal record showing all information about one item completely together
Rule	Complete instruction pairing selector with property and value for styling
SELECT	Command to fetch chosen columns from a table for viewing results
Script	List of instructions running inside browser to make page interactive for users
Search	Looking through array to find element matching desired value quickly
Selector	Pattern choosing which HTML elements to style together as group
Sensor	Device detecting temperature light or motion and sending data to system
Server	Powerful computer storing many files for many users together
Set	Collection holding only unique values with no duplicates allowed automatically for uniqueness
Signature	First line showing function name and its expected parameters clearly together
Size	Total number of elements currently stored inside the array altogether
Slice	Part of list taken as new smaller list like first three items
Smart Device	Object that senses acts and connects to internet automatically without human help
Sort	Arranging list elements in order like smallest to largest automatically quickly
Sorting	Arranging items in order like smallest to largest for easy searching later
Stack	Memory storing each waiting function call until base case reached for completion
String	Sequence of characters like name or sentence inside quotes together
Substring	Part of a string taken from middle like first three letters
Superposition	Being in multiple states simultaneously until measured for quantum advantage greatly
Tag	Word inside angle brackets marking start and end of element like <p>
Target	Value you are trying to locate inside the collection of data currently
Training	Adjusting connections by showing many examples until accuracy improves gradually step
Traverse	Visiting each element of array one by one in order sequentially
Try	Block where you attempt code that might cause an error safely
Tuple	Ordered collection that cannot be changed after creation keeping data safe permanently
Union	Combining all elements from two sets together without duplicating any value
Unique	Appearing only once with no repetition inside the collection for clarity
Update	Changing value already stored under existing key to new value
Upload	Sending your file from device to cloud storage
Value	Result produced by function after doing its calculation correctly for use
WHERE	Clause filtering rows to show only those meeting a condition for query
Write	Saving data from program into file for future use permanently

Self-Assessment

Chapter	I Can	With Help	Not Yet
Advanced Programming Concepts	☐	☐	☐
Data Structures	☐	☐	☐
Database Fundamentals	☐	☐	☐
Web Technologies	☐	☐	☐
Algorithms and Computational Thinking	☐	☐	☐
Emerging Technologies	☐	☐	☐